SecureLink

Project Synopsis Submitted

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by

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Objective:

The objective of this project is to design and implement a secure, role-based company intranet that allows authenticated employees to exchange text, documents, and images while preventing unauthorized external access. The system aims to address the risks of uncontrolled internal communication and unverified file sharing by providing a three-layered solution:

1. Secure Access Control: Company-managed user accounts with hashed passwords, role-based permissions, and a two-device active session limit.
2. Validated Communication & File Sharing: Internal text messaging along with document and image sharing, backed by file-type validation and cryptographic (SHA-256) integrity verification.
3. Network Security Analysis: Capture and analysis of intranet traffic using Wireshark and PyShark, with extracted packet metadata stored in SQLite for structured auditing via a security dashboard.

Through this project, we intend to build a usable intranet application that also functions as a cybersecurity analysis platform, combining authentication, cryptography, and network-traffic analysis to protect internal company communication.

Scope:

This project focuses on building a private, company-only intranet limited to text messaging and document/image sharing between authenticated employees within a controlled network. The work covers user authentication, role-based access control, a two-device session limit, file extension/MIME/signature validation, SHA-256 integrity verification, and audit logging. A responsive web interface will allow the same application to be used across PC and mobile devices without a separate native app. The network security component will capture live intranet traffic with Wireshark, analyze it programmatically with PyShark, and convert the extracted packet metadata into SQLite for review through a security dashboard. Image-specific metrics such as entropy, MSE, PSNR, NPCR, and UACI will be treated as an optional research/evaluation layer rather than core product functionality.

Need for the Application:

Organizations frequently need to exchange internal documents, images, and messages between employees without exposing company information to external users or uncontrolled public platforms. General-purpose tools such as personal email or public chat apps do not guarantee company-only access, verifiable file integrity, or accountability for user actions, and offer no way to audit or analyze the underlying network traffic. This project addresses that need by combining strong authentication, role-based authorization, validated file sharing, and cryptographic integrity checks with a dedicated traffic-analysis layer, ensuring that sensitive company communication remains confidential, tamper-evident, and traceable.

Project Description:

Problem Statement:

Companies need a way to let employees exchange internal messages, documents, and images without risking exposure to unauthorized users or uncontrolled external platforms. Without proper safeguards, shared files can be silently modified during transfer or storage, disguised as a permitted file type, or accessed by users outside their assigned role. In addition, most everyday collaboration tools provide little visibility into the underlying network traffic, making it hard to detect suspicious activity or verify that security controls are actually working. There is therefore a need for an intranet platform that enforces

authentication, authorization, file validation, and integrity checking, while also allowing its own network traffic to be captured and analyzed for security assurance.

Proposed Solution:

The proposed solution is a secure company intranet where an administrator creates company user profiles with assigned roles (e.g. Administrator, HR, Finance, Developer). Users authenticate with securely hashed passwords and are limited to two active device sessions, each with expiration and inactivity timeouts. Authenticated and authorized users can exchange text messages and share documents/images, with every uploaded file checked against its extension, MIME type, and binary signature before being accepted. Each accepted file is hashed with SHA-256, transmitted over TLS, and re-verified after receipt so that any tampering is detected and logged. Every security-relevant action is recorded in an audit log. A responsive web interface makes the same application usable from both desktop and mobile devices. In parallel, intranet traffic is captured with Wireshark, processed programmatically with PyShark, and converted into a SQLite database that feeds a security dashboard summarizing users, sessions, files, and network events.

Functionalities to be Implemented:

- Company-created user accounts with configurable roles and hashed password storage.
- Login/logout with a two-device active session limit, session expiration, and inactivity timeout.
- Role-based authorization restricting access to permitted resources only.
- Internal text messaging between authenticated company users.
- Document and image sharing with file extension, MIME type, and signature (magic byte) validation.
- SHA-256 file hashing with post-transfer integrity verification.
- Audit logging of logins, file transfers, access denials, and security events.
- Responsive web interface usable across PC and mobile devices.
- Network traffic capture (Wireshark) and programmatic analysis (PyShark) of intranet activity.
- PCAP-to-SQLite pipeline feeding an administrator security dashboard.

Hardware Requirements:

- Processor: Intel Core i5 (8th generation or equivalent AMD Ryzen 5) or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 20 GB available SSD/HDD space (50 GB+ SSD recommended)
- Network: Ethernet or Wi-Fi connection to the company/local intranet
- Server: A dedicated PC, laptop, or virtual machine to run the backend and database
- Client Devices: At least 2 devices for testing (PC/Laptop and Smartphone/Tablet)
- GPU: Not required, as the system does not involve deep-learning model training

Software Requirements:

- Operating System: Ubuntu Linux 22.04 LTS (server, recommended) / Windows 10/11; client access via Windows, Linux, macOS, Android, or iOS through a responsive web app
- Programming Languages: Python (backend, cryptography), JavaScript/TypeScript (frontend)
- Frameworks & Libraries: FastAPI/Flask, React, PyNaCl/cryptography (AES-GCM), Argon2id/bcrypt, PyShark
- Database: PostgreSQL or SQLite
- Tools: Wireshark (traffic capture), Git (version control), Docker (containerization, optional)

- Web Browser: Chrome/Firefox for accessing the web interface

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